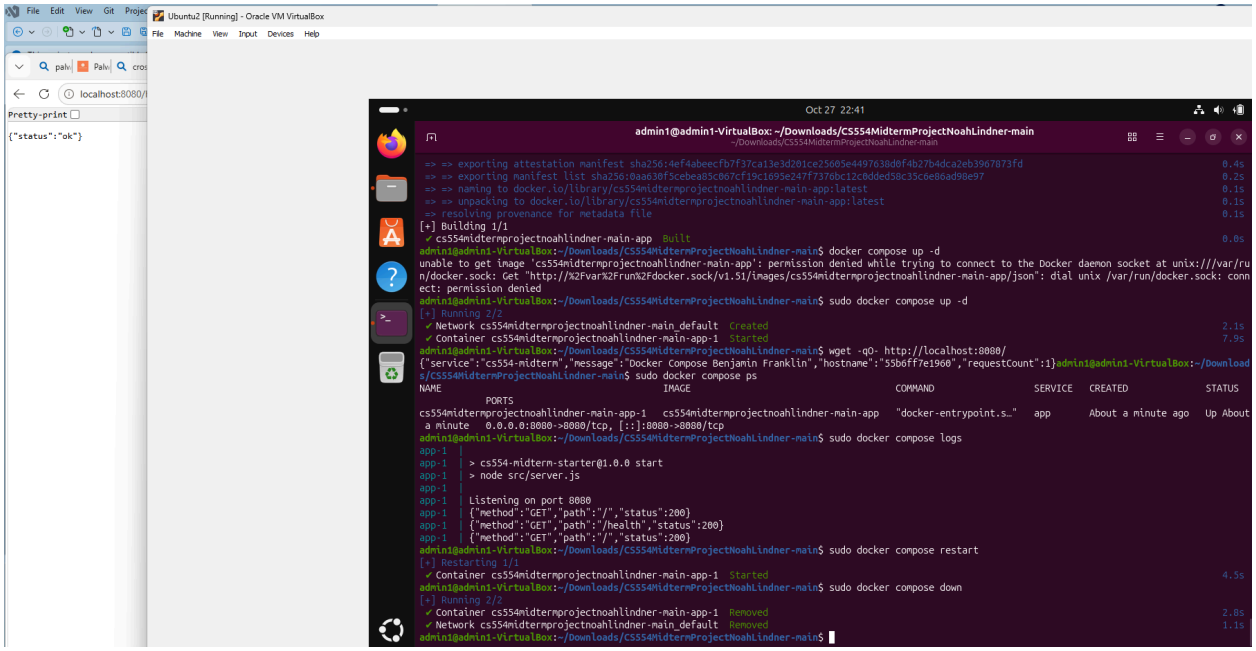
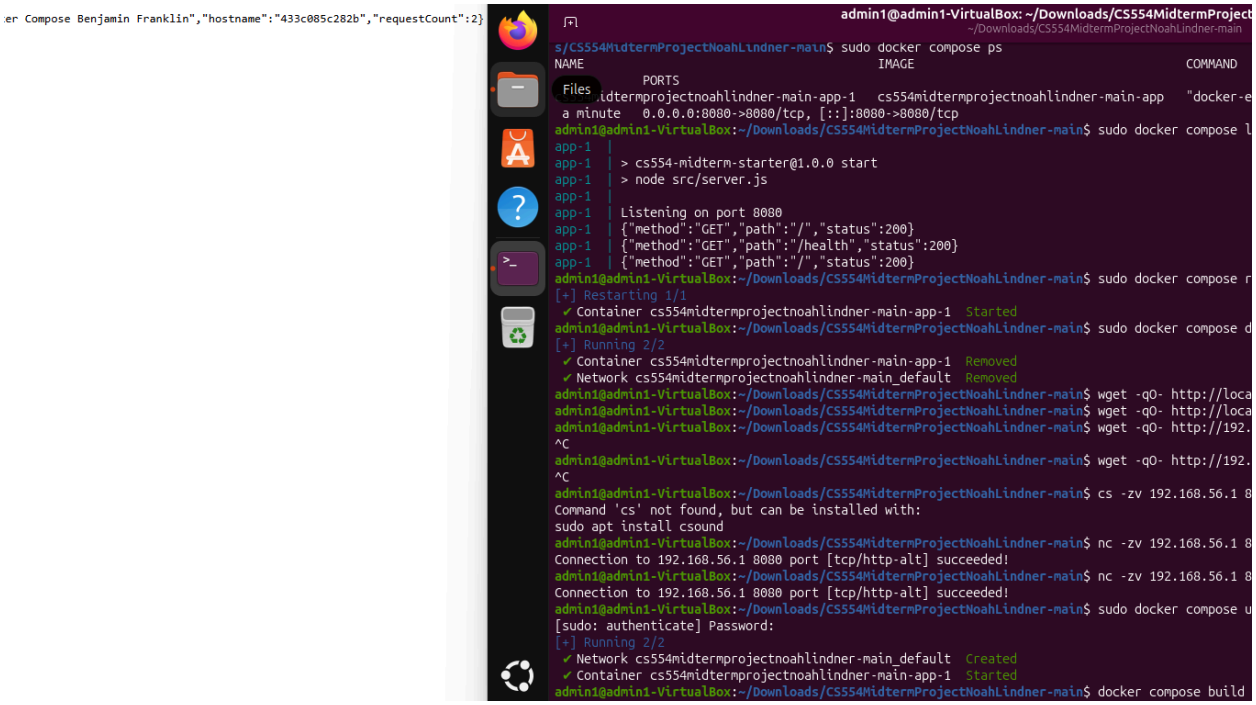
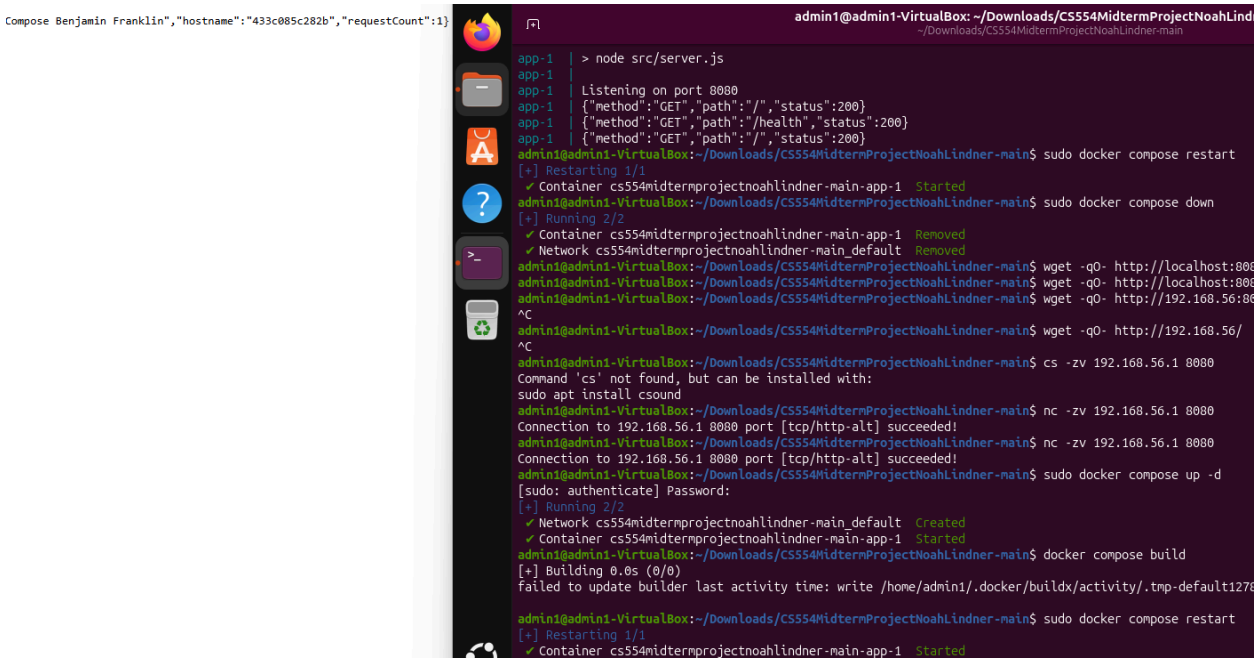
The entirety of part 2 cmd build and host connection:



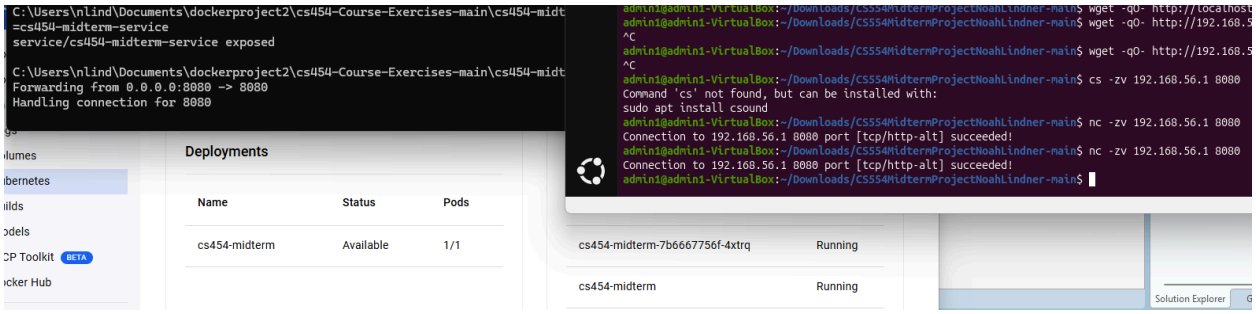
Part 2: before restart-request count = 2



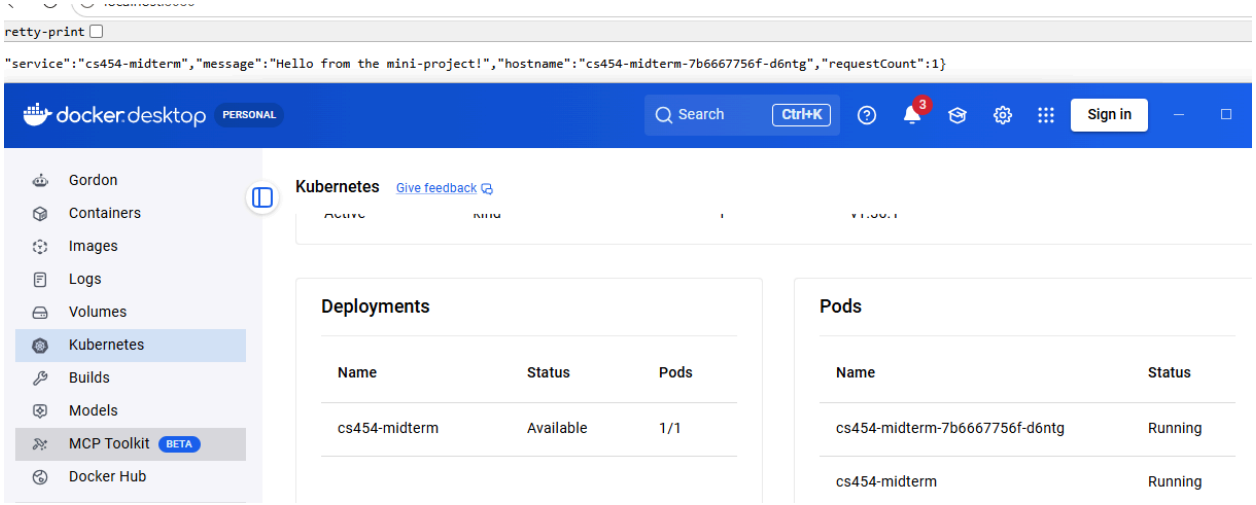
Part 2: after restart request count = 1



Part 3 kubernetes host -> VM connection proof



Part 3 kubernetes functioning app proof



Part 3: kubernetes deletion and auto replacement

cs454-midterm-7b6667756f-d6ntg	Running

Name	Status
cs454-midterm-7b6667756f-z9rmg	Running

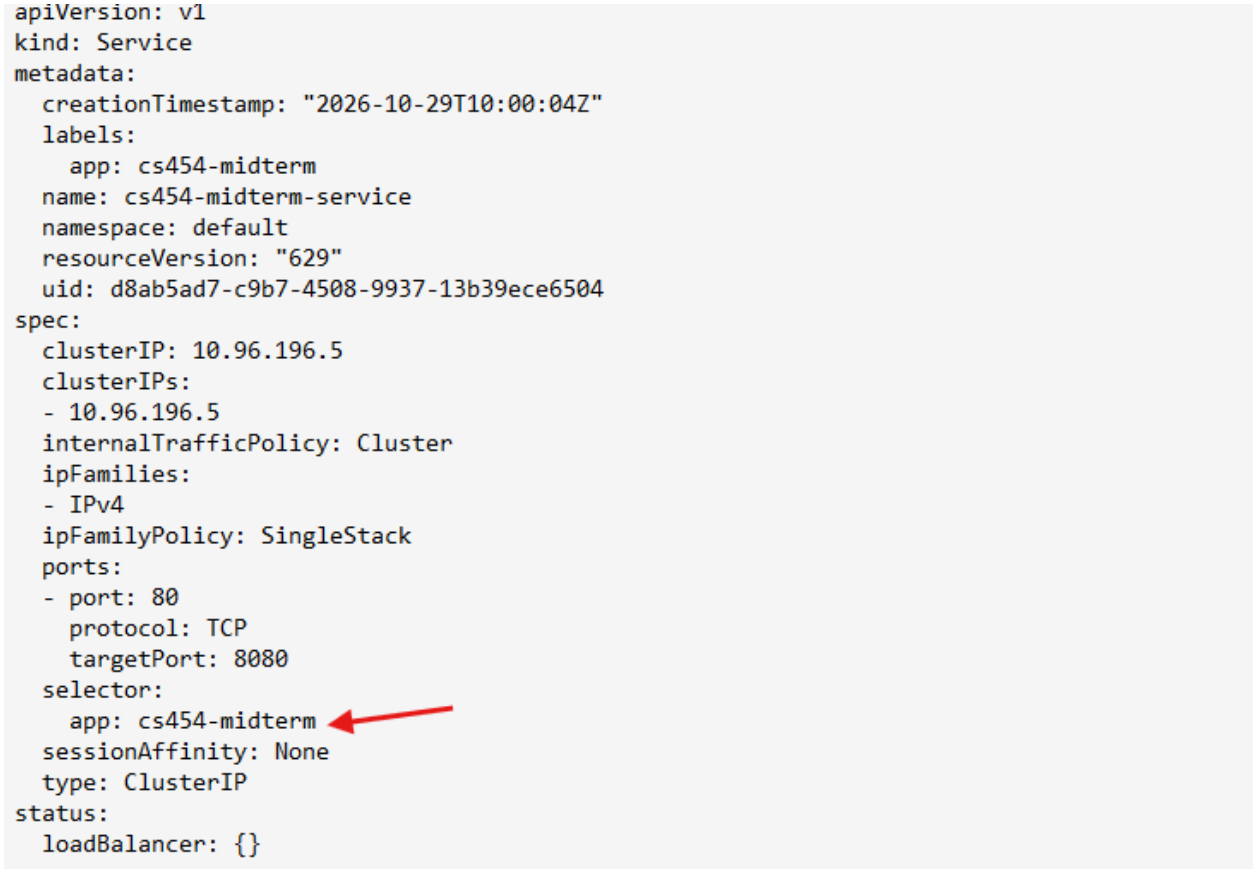
```
C:\Users\nlind>kubectl delete pod cs454-midterm-dfcd6f5f5-tkdjb
pod "cs454-midterm-dfcd6f5f5-tkdjb" deleted from default namespace

C:\Users\nlind>kubectl get pods
NAME                              READY   STATUS    RESTARTS      AGE
cs454-midterm-dfcd6f5f5-4rw47   1/2     Error    2 (17s ago)   22s
cs454-midterm-dfcd6f5f5-rzlxq   1/2     Error    1 (11s ago)   13s

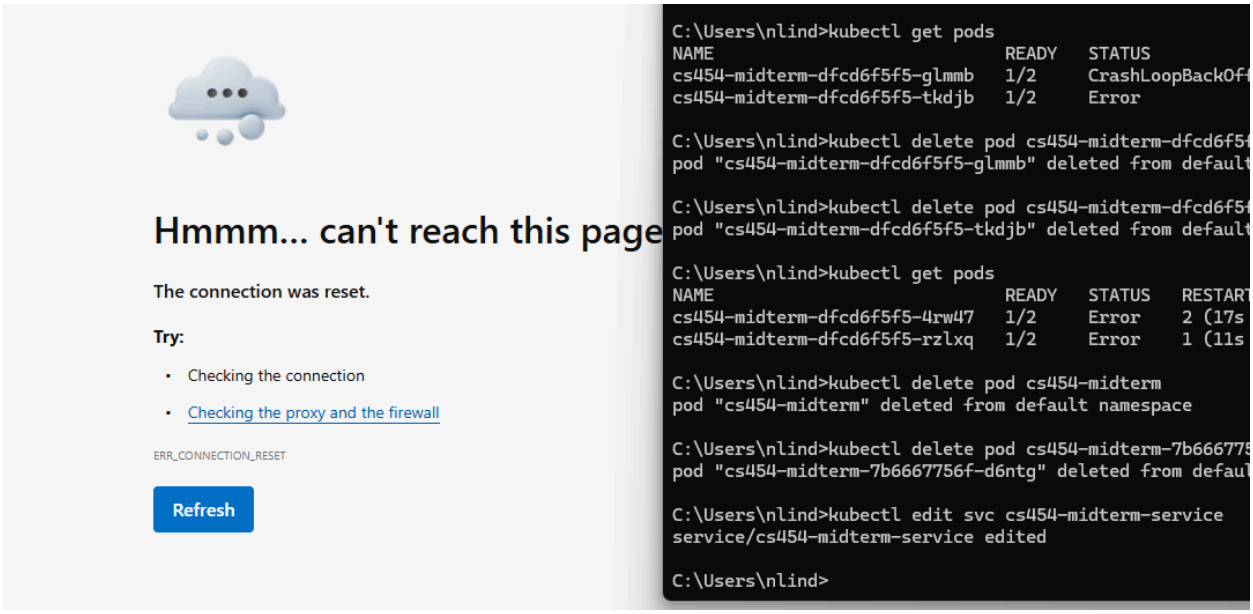
C:\Users\nlind>kubectl delete pod cs454-midterm
pod "cs454-midterm" deleted from default namespace

C:\Users\nlind>kubectl delete pod cs454-midterm-7b6667756f-d6ntg
pod "cs454-midterm-7b6667756f-d6ntg" deleted from default namespace
```

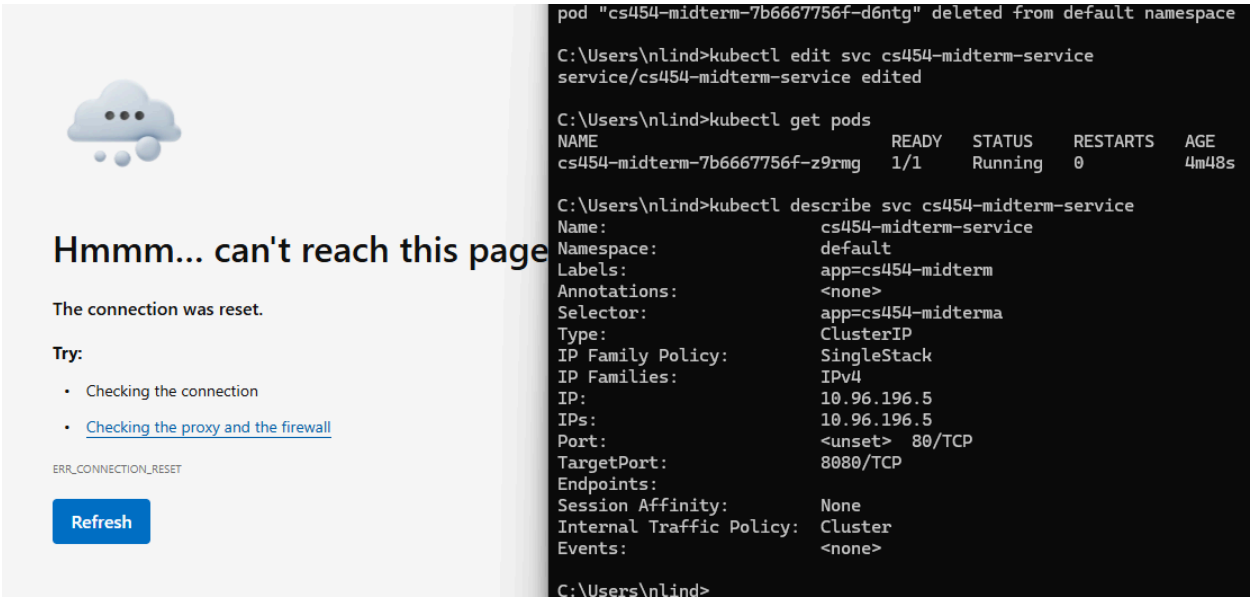
Part4: selector app address changing



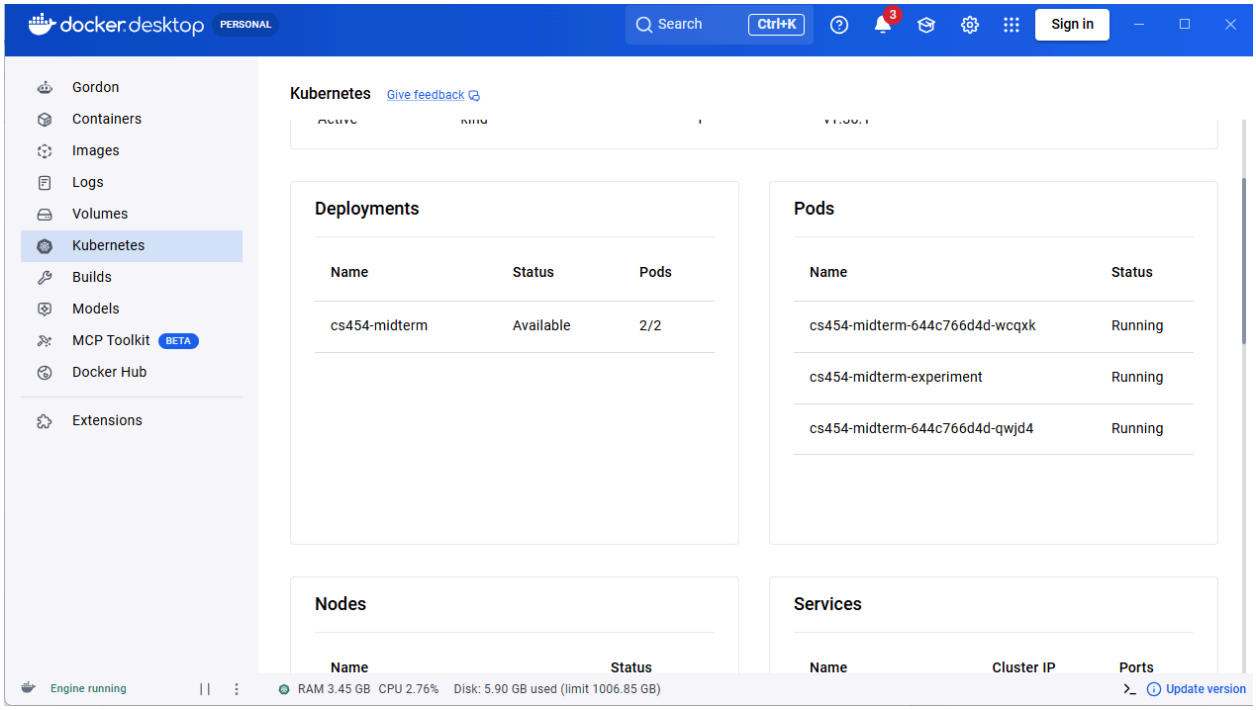
```
apiVersion: v1
kind: Service
metadata:
  creationTimestamp: "2026-10-29T10:00:04Z"
  labels:
    app: cs454-midterm
  name: cs454-midterm-service
  namespace: default
  resourceVersion: "629"
  uid: d8ab5ad7-c9b7-4508-9937-13b39ece6504
spec:
  clusterIP: 10.96.196.5
  clusterIPs:
  - 10.96.196.5
  internalTrafficPolicy: Cluster
  ipFamilies:
  - IPv4
  ipFamilyPolicy: SingleStack
  ports:
  - port: 80
    protocol: TCP
    targetPort: 8080
  selector:
    app: cs454-midterm
  sessionAffinity: None
  type: ClusterIP
status:
  loadBalancer: {}
```



Part4: pod data



Readiness failure and recovery pod proof:




```
C:\Users\nlind>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
cs454-midterm-644c766d4d-qwjd4      1/1     Running   0          102s
cs454-midterm-644c766d4d-wcqxk      1/1     Running   0          3m54s
cs454-midterm-experiment            0/1     Running   0          2m7s

C:\Users\nlind>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
cs454-midterm-644c766d4d-qwjd4      1/1     Running   0          111s
cs454-midterm-644c766d4d-wcqxk      1/1     Running   0          4m3s
cs454-midterm-experiment            1/1     Running   0          2m16s
```